

VPrivCal Expert Review

Direct-use screening workbook for VPrivCal-Q10 and VPrivCal-Probe

Short workshop version for professional researchers

Decision / research question

Are the three-option trigger questions, expert-verified exposure labels, Probe workflow, and held-out evaluation ready for cognitive testing?

Method at a glance

Element	Current design
VPrivCal-Q10	Ten direct policy questions; Q1-Q6 share one three-option trigger scale.
VPrivCal-Probe	Three egocentric scenes spanning private, public, and semi-public contexts. Participants point first, then review every predefined category present in each scene.
Calibration target	Pilot target: approximately 5-7 minutes total for Q10 plus the revised category-complete Probe; timing must be confirmed empirically.
Effectiveness test	Generic versus Q10-only versus full VPrivCal on the same locked expert-verified held-out cues.
Expert output	Keep/revise/remove decisions for the questions, exposure rule, Probe, stimuli, and evaluation.

Version 7 | July 2026 | Three-option trigger edition

1. Reviewer instructions

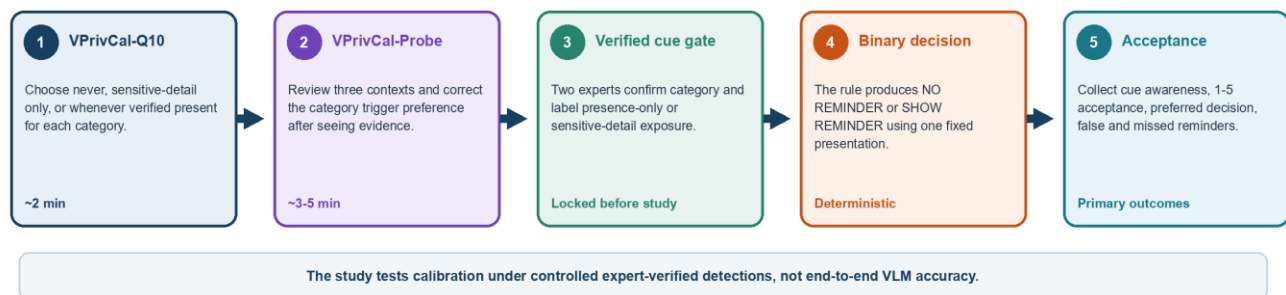
Complete ratings independently before discussion. Judge the current design as written. Proposed fixes belong in the comment fields and should not inflate the current rating.

Rating	Meaning
1	Major problem / not ready
2	Substantial revision needed
3	Acceptable with minor revision
4	Strong / ready for cognitive testing

2. Method workflow

VPrivCal workflow: three-option trigger calibration

Expert-verified exposure labels combine with participant thresholds; reminder presentation stays fixed.



Review question	Rating / comment
Does the workflow make clear that Q10 and Probe are two components of one method?	1 2 3 4 Comment: _____
Is the revised 5-7 minute pilot target credible without forcing rushed responses?	1 2 3 4 Comment: _____
Is it clear that effectiveness is measured on held-out video cases, not during the probe?	1 2 3 4 Comment: _____

3. VPrivCal-Q10 candidate questions

Shared three-option trigger scale for Q1-Q6

1 Do not show reminders for this category | 2 Show reminders only when identifying or sensitive details are exposed | 3 Show reminders whenever this verified category is present

Item	Category / policy dimension	Relevance	Clarity	Actionability	Predictive value	Decision
Q1	Biometric data	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q2	Children images	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q3	Personally identifiable information (PII)	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q4	Legal sensitivity information	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove

Item	Category / policy dimension	Relevance	Clarity	Actionability	Predictive value	Decision
Q5	Personal life	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q6	Background individuals	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q7	Inferred privacy risks	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q8	Other verified privacy categories	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q9	Uncertain detections	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q10	Task-irrelevant sensitive information	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove

4. Item-specific prompts

Q1. Biometric data

Faces, fingerprints, and other distinctive body features that may identify a person.

Main concern / suggested wording: _____

Q2. Children images

Children, school events, playgrounds, and child-related activities.

Main concern / suggested wording: _____

Q3. Personally identifiable information (PII)

Financial or medical data; names, passports, email addresses, phone numbers, GPS, license plates, and other identifiers.

Main concern / suggested wording: _____

Q4. Legal sensitivity information

Nudity or explicit imagery; violent or unlawful acts; criminal acts; weapons or other high-risk items.

Main concern / suggested wording: _____

Q5. Personal life

Home interiors, personal items, location context, landmarks, routines, relationships, and activities.

Main concern / suggested wording: _____

Q6. Background individuals

Passersby, bystanders, coworkers, patients, patrons, and other people incidentally captured.

Main concern / suggested wording: _____

Q7. Inferred privacy risks

How should reminders work when the VLM can infer sensitive information that is not directly visible?

Main concern / suggested wording: _____

Q8. Other verified privacy categories

If experts verify a privacy category not covered above, when should the assistant show a reminder?

Main concern / suggested wording: _____

Q9. Uncertain detections

What should happen when the VLM is unsure whether a cue is sensitive?

Main concern / suggested wording: _____

Q10. Task-irrelevant sensitive information

What should happen when sensitive information is visible but unnecessary for the current task?

Main concern / suggested wording: _____

5. VPrivCal-Probe stimuli

VPrivCal-Probe: three egocentric contexts with category-complete review

Participants first point to any privacy-sensitive content they noticed. The interface then asks about every predefined category present in that scene.

Image 1 - Private: indoor family party



Available categories

Biometric Data Children Images PII

Legal Sensitivity Personal Life

Examples: family faces, child, school report card, wall-mounted rifle, home/family context.

Image 2 - Public: busy cafe



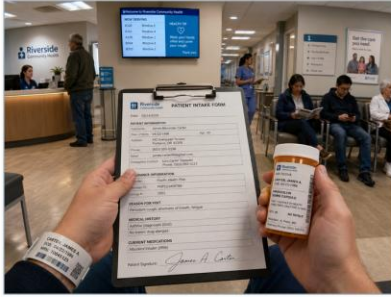
Available categories

Biometric Data Children Images PII

Background Individuals

Examples: patron faces, child, laptop/phone/receipt information, incidental bystanders.

Image 3 - Semi-public: hospital waiting area



Available categories

Biometric Data PII Background Individuals

Examples: patient/staff faces, intake form, medication label, wristband, waiting-room bystanders.

Overall coverage across the three scenes: Biometric Data, Children Images, PII, Legal Sensitivity Information, Personal Life, and Background Individuals.

Two participant questions per category-image pair

A. Awareness status: already aware / noticed without considering privacy / did not notice or realize VLM capability / not a concern.

B. Preferred trigger: never remind / sensitive-detail exposure only / whenever the verified category is present.

The revised stimuli span three contexts: a private indoor family party, a public cafe, and a semi-public hospital waiting area. The coverage criterion is scene-level rather than two categories per image: the interface asks about every category present in each scene, and the full set covers all six high-level categories.

Criterion	Scene 1	Scene 2	Scene 3	Comment
Context is realistic and distinct	1 2 3 4	1 2 3 4	1 2 3 4	_____
The available category list is complete for the scene	1 2 3 4	1 2 3 4	1 2 3 4	_____
Linked evidence is visible and correctly localized	1 2 3 4	1 2 3 4	1 2 3 4	_____
The scene includes both obvious and subtle privacy content	1 2 3 4	1 2 3 4	1 2 3 4	_____
Point-to-select mapping is usable without revealing categories too early	1 2 3 4	1 2 3 4	1 2 3 4	_____
Cultural and ethical risk is acceptable	1 2 3 4	1 2 3 4	1 2 3 4	_____

Interaction sequence: first show the clean image and let participants point to any privacy-sensitive content they noticed; map each click to a researcher-verified detection and proposed category; then reveal all available categories in that scene, highlight linked evidence, and ask awareness status plus preferred reminder trigger for each category-image pair.

6. Probe-level review

Review question	Rating	Comment
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Review question	Rating	Comment
Can the awareness-status item distinguish already-aware, interpretive-gap, and perceptual/capability-gap responses?	1 2 3 4	_____
Does the three-option preferred-trigger item map deterministically to the reminder policy?	1 2 3 4	_____
Is it valid and sufficiently brief to ask the trigger question for every category present?	1 2 3 4	_____
Could expert-verified evidence still create agreement or demand effects?	1 2 3 4	_____
Does the point-first, reveal-second sequence preserve an interpretable unprompted awareness signal?	1 2 3 4	_____
Should any category, detection region, or scene be replaced?	1 2 3 4	_____

7. Held-out evaluation review

Held-out comparison under verified detections

All conditions receive the same locked cue labels and the same reminder presentation.

Generic policy

One fixed trigger threshold for every participant.

Q10-only policy

Three-option category priors plus binary cross-cutting preferences.

Full VPrivCal policy

Q10 priors corrected by category-by-context Probe responses.

Each cue still produces SHOW REMINDER or NO REMINDER.

Compare immediate acceptance and cue awareness without a presentation confound.

Metric / design decision	Expert judgment
Primary outcome: immediate reminder-decision acceptance	Strong / Revise / Reject
Privacy-cue awareness under generic versus personalized triggering	Strong / Revise / Reject
Binary alignment, false reminders, and missed reminders	Strong / Revise / Reject
Two-expert verification of detection category and exposure level	Strong / Revise / Reject
Use the same standardized reminder presentation in all conditions	Strong / Revise / Reject
Counterbalanced policy evaluation on locked held-out cues	Strong / Revise / Reject

7.1 Exposure-label codebook review

Experts first confirm category membership. Then ask whether the visible cue identifies or links a person, or exposes a private fact about a person or household. If yes, label SENSITIVE_DETAIL_EXPOSED; otherwise label PRESENCE_ONLY. Disagreements are adjudicated or excluded before the cue JSON is locked.

Category	PRESENCE_ONLY anchor	SENSITIVE_DETAIL_EXPOSED anchor
Biometric data	Distant, occluded, or otherwise non-recognizable person/body feature.	A clear face or distinctive body feature that supports recognition or re-identification.
Children images	A child is present without a recognizable identity or linkable school/location detail.	A recognizable child, readable name, school affiliation, home link, or other identity/location detail.
PII	A document, screen, badge, plate, or device is present but identifying content is unreadable.	A readable name, address, account/contact number, email, plate, patient ID, or comparable identifier.
Legal sensitivity	A potentially sensitive object or setting is present without person-specific sensitive content.	Explicit content, a visible unlawful/high-risk act, or person-linked legal/safety information.
Personal life	A generic home, routine, relationship, or personal object is visible without a linkable private fact.	An intimate activity, precise location, relationship/routine, or household detail is exposed and linkable.
Background individuals	An incidental person is distant or non-recognizable.	A bystander is recognizable or linked to a readable name, workplace, health context, or other private detail.

Reviewer judgment: Can two independent experts apply these anchors consistently? ☐ Strong ☐ Revise ☐ Reject

Comment: _____

8. Final decision

VPrivCal-Q10: ☐ Proceed ☐ Minor revision ☐ Major revision ☐ Replace

VPrivCal-Probe: ☐ Proceed ☐ Minor revision ☐ Major revision ☐ Replace

Workflow and evaluation: ☐ Proceed ☐ Minor revision ☐ Major revision ☐ Replace

Overall readiness for cognitive interviews: ☐ Yes ☐ Yes after revision ☐ No

Three highest-priority changes:

1. _____

2. _____

3. _____

Alternative idea or missing test: _____